

Program 1 — Welcome Message Write a program that asks the user for their name and prints a welcome message. For example if the user enters "Kruthi" the program should print "Hello Kruthi, welcome!"

Program 2 — Pay Calculator Write a program to prompt the user for hours and rate per hour using input to compute gross pay. Use 35 hours and a rate of 2.75 per hour to test the program (the pay should be 96.25). You should use input to read a string and float() to convert the string to a number. Do not worry about error checking or bad user data.

Chapter 3 — Conditional Code:

Program 3 — Overtime Pay Rewrite your pay program using a function called computepay that takes two parameters (hours and rate). If the hours worked exceeds 40 then compute pay with overtime at 1.5 times the hourly rate for all hours over 40. Use 45 hours and a rate of 10.50 per hour to test the program (the pay should be 498.75). You should use input to read a string and float() to convert the string to a number. Do not worry about error checking or bad user data.

Program 4 — Grade Checker Write a program to prompt the user for a score between 0.0 and 1.0. If the score is out of range print an error. If the score is between 0.9 and 1.0 print A. If the score is between 0.8 and 0.9 print B. If the score is between 0.7 and 0.8 print C. If the score is between 0.6 and 0.7 print D. If the score is below 0.6 print F. Test the program with these inputs: 0.95, 0.85, 0.75, 0.65, 0.55, 1.1, -0.1.

Chapter 4 — Functions:

Program 5 — Celsius to Fahrenheit Write a function called convertTemp that takes a temperature in Celsius as a parameter and returns the temperature in Fahrenheit. The formula is $F = (C \times 1.8) + 32$. Ask the user to enter a temperature in Celsius, call your function, and print the result. Test with 0 Celsius (should give 32 Fahrenheit) and 100 Celsius (should give 212 Fahrenheit).

Program 6 — computepay with function Rewrite your pay computation with overtime using a function called computepay that takes two parameters hours and rate. The function should return the pay. Use input to prompt the user for hours and rate. Test with 45 hours at 10.50 per hour (answer should be 498.75).

Chapter 5 — Loops:

Program 7 — Count to 10 Write a program that uses a loop to print the numbers 1 through 10 one per line.

Program 8 — Sum of a list Write a program that adds up the numbers in this list and prints the total: [3, 41, 12, 9, 74, 15] Do not use sum(). Use a loop and add each number manually.

Program 9 — Find the largest Write a program that finds and prints the largest number in this list: [3, 41, 12, 9, 74, 15] Do not use max(). Use a loop and track the largest number yourself.

Program 10 — Find the smallest Write a program that finds and prints the smallest number in this list: [3, 41, 12, 9, 74, 15] Do not use min(). Use a loop and track the smallest number yourself.

Program 11 — Running total Write a program that repeatedly prompts the user to enter numbers. When the user types "done" instead of a number the program should print the total, the count of how many numbers were entered, and the average. If the user enters something that is not a number and is not "done" print an error message and ask again. Test by entering: 3, 9, done (total should be 12, count should be 2, average should be 6.0).

Program 12 — Guessing game Write a program that picks a secret number between 1 and 10. Ask the user to guess the number. If the guess is too low print "Too low, try again". If the guess is too high print "Too high, try again". Keep asking until they guess correctly then print "Well done, you got it!" and tell them how many guesses it took.